

# Conjecture I: Xenon Recoil as a Constraint on Cold SI-Membrane Relics

Clarification to Opus III and Opening Direction for Opus IV

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## Abstract

Opus III formulated polarized SI-membrane cosmology around a universal matter/conjugate orientation label,  $\chi$ , whose branch separation occurs through condensation of the SI order parameter. This note adds a clarification and a first phenomenological conjecture.

The clarification concerns the role of the approximate 40 MeV scale. The universal label  $\chi$  does not by itself set the final baryon abundance. Rather,  $\chi$  identifies what is being separated, while the SI transition temperature  $T_c$  determines when separation becomes irreversible. Before SI condensation, matter-oriented and conjugate-oriented components remain in a common annihilation history. When SI condenses, cross-orientation annihilation is interrupted by branch separation. Therefore the observed baryon-to-photon ratio,  $\eta_B \sim 6 \times 10^{-10}$ , provides the primary empirical clock for estimating  $T_c \sim 38\text{--}40$  MeV.

The conjecture concerns the neutral relic sector  $R_{\text{SI}}$ . If the SI transition leaves a cold, neutral, nonrelativistic residue, then rare nuclear recoils in xenon-type detectors may eventually provide a laboratory constraint on that residue. This is not a claim that current xenon data prove dark matter or SI relics. The recent LUX-ZEPLIN result consists of one event consistent with a  $248 \pm 23$  (stat)  $\pm 23$  (sys) keV nuclear recoil, with a global significance of  $2.6\sigma$ , and therefore remains far below discovery level. It is treated here only as motivation for a possible signature class.

The central constraint is relic statistics. If  $R_{\text{SI}}$  is the dark matter, then  $m_R n_R \approx 5.3 m_p n_b$ . Thus the SI program must derive the relic-to-baryon number ratio  $\kappa_R = n_R/n_b$ . A first sanity check shows that a naive one-scale scalar condensate with  $m_\phi \sim \phi_0 \sim T_c \sim 40$  MeV overproduces dark matter by roughly  $10^6\text{--}10^7$ . Therefore the simplest one-scale relic version is disfavored. A viable relic sector must involve a suppressed amplitude, a lighter SI oscillation mode, a compact relic population with calculable number density, or another mechanism that decouples relic statistics from the transition scale without making them arbitrary.

## 1 Status of the note

This note is not Opus IV. It is Conjecture I: a short phenomenological bridge between Opus III and the first possible experimental constraints on the SI relic sector.

Opus I introduced polarized membrane cosmology as a three-sector yield problem: a visible matter-oriented branch  $A$ , a conjugate branch  $B$ , and a neutral relic sector associated with the branch-separating order parameter. Opus II named the membrane SI, emphasized universal orientation sorting, and kept  $\Omega_{\text{DM}}/\Omega_b \approx 5.3$  as the first quantitative target of the relic channel.

Opus III added the central inversion: the matter/conjugate label is not primarily a property of individual particles. It is a universal pre-particle orientation label, while observed particles are quantized projection modes of a deeper excitation.

Conjecture I adds two items. First, it clarifies the role of the 40 MeV transition scale so that the scale is not decorative. Second, it asks whether a cold neutral SI relic could, in principle, produce rare xenon nuclear recoils. The conjecture is deliberately modest. It does not claim detection. It defines a constraint.

## 2 Clarification to Opus III: why 40 MeV matters

The universal orientation label  $\chi$  should not be confused with a preloaded final baryon abundance.

$$\chi = \text{the label telling SI what orientation is being separated,} \quad (1)$$

$$T_c = \text{the epoch when separation becomes irreversible,} \quad (2)$$

$$\eta_B = \text{the residue left when annihilation is interrupted.} \quad (3)$$

*Reading:* The label  $\chi$  answers “what is being separated.” The temperature  $T_c$  answers “when does separation freeze.” The observed matter/photon ratio answers “how much remained when that happened.”

Before SI condensation, matter-oriented and conjugate-oriented components remain in a shared effective thermal history. Ordinary annihilation continues. The symmetric matter-antimatter population is progressively converted into radiation.

When the SI order parameter condenses, the remaining matter-oriented and conjugate-oriented populations become dynamically separated into branch sectors. After this point, cross-orientation annihilation no longer continues inside branch  $A$ .

Thus the visible residue is not fixed by  $\chi$  alone. It is fixed by how much oriented matter remains when SI condensation occurs.

The observed baryon-to-photon ratio,

$$\eta_B \sim 6 \times 10^{-10}, \quad (4)$$

therefore acts as a clock. It says that annihilation had already proceeded almost to completion before branch separation, but not so far that the surviving branch-visible residue was erased below the observed value.

In plain terms, the membrane does not create the matter excess from nothing. The membrane stops the annihilation game when the remaining residue matches what we observe. This gives the approximate estimate

$$T_c \sim 38\text{--}40 \text{ MeV}. \quad (5)$$

A schematic expression for the intended relation is

$$\eta_A = F(\Gamma_{\text{ann}}, H, T_c, \mathcal{S}_{\text{SI}}), \quad (6)$$

where  $\Gamma_{\text{ann}}$  is the annihilation history,  $H$  is the expansion rate, and  $\mathcal{S}_{\text{SI}}$  is the SI branch-separation operation.

*Reading:* The baryon/photon ratio estimates the timing of the SI transition. The dark-to-baryonic ratio constrains the relic yield of that transition. These are related outputs of one event, but they are not the same equation.

### 3 The relic sector cannot be “floating energy”

The relic sector  $R_{\text{SI}}$  cannot be described merely as leftover or unattached energy. That language is too loose because it does not specify the equation of state.

If  $R_{\text{SI}}$  is radiation-like, then

$$\rho_R \propto a^{-4}. \quad (7)$$

It redshifts like photons. It does not behave as cold dark matter. It would instead be dark radiation.

That possibility is strongly constrained. Planck 2018, combined with BAO, gives  $N_{\text{eff}} = 2.99 \pm 0.17$ , consistent with the Standard Model neutrino expectation. The same Planck results give  $\Omega_c h^2 \simeq 0.120$  and  $\Omega_b h^2 \simeq 0.0224$ , implying  $\Omega_c/\Omega_b \simeq 5.3$ .

Therefore the allowed relic is not

$$R_{\text{SI}} = \text{dark radiation}. \quad (8)$$

The viable relic must be cold and matter-like:

$$R_{\text{SI}} = \text{cold neutral SI residue}. \quad (9)$$

This residue may be a massive neutral excitation, a compact soliton, a nonrelativistic condensate, or another stable order-parameter remnant.

The energy is not floating like light. It is frozen into a cold neutral form that gravitates like matter.

### 4 Relic statistics

The dark matter abundance is known well enough to impose a hard accounting constraint. If the SI relic sector is the dark matter, then

$$\rho_R \approx 5.3 \rho_b. \quad (10)$$

Since

$$\rho_R = m_R n_R, \quad \rho_b \approx m_p n_b, \quad (11)$$

the relic sector must satisfy

$$m_R n_R \approx 5.3 m_p n_b. \quad (12)$$

Equivalently,

$$\boxed{\frac{m_R}{m_p} \approx 5.3 \frac{n_b}{n_R}.} \quad (13)$$

Define the relic-to-baryon number ratio

$$\kappa_R \equiv \frac{n_R}{n_b}. \quad (14)$$

Then the relic mass is

$$\boxed{m_R \approx \frac{5.3}{\kappa_R} m_p.} \quad (15)$$

*Reading:* This equation is not the solution. It is the constraint. The program must derive  $\kappa_R$ . Once  $\kappa_R$  is known, the relic mass is no longer free.

If  $\kappa_R \sim 1$ , the relic mass is a few GeV. If detector recoil kinematics require a much heavier relic, then  $\kappa_R \ll 1$ , meaning SI relics must be much rarer than baryons. If  $\kappa_R \gg 1$ , the relic mass must be much lighter.

Thus the SI program cannot choose relic mass and relic number density independently. A predictive version must derive  $\kappa_R$  from SI projection structure, index multiplicities, compact relic statistics, or another calculable feature of the transition.

## 5 First sanity check: scalar condensate

The simplest cold candidate is an oscillating scalar condensate of the SI order parameter  $\phi_{\text{SI}}$ .

For a scalar condensate,

$$\rho_\phi \sim \frac{1}{2} m_\phi^2 \phi_0^2, \quad (16)$$

where  $m_\phi$  is the oscillation mass and  $\phi_0$  is the initial displacement or residual amplitude.

A naive one-scale version would set both quantities by the SI transition scale:

$$m_\phi \sim \phi_0 \sim T_c \sim 40 \text{ MeV} = 0.04 \text{ GeV}. \quad (17)$$

Then

$$\rho_\phi \sim \frac{1}{2} (0.04)^4 \sim 10^{-6} \text{ GeV}^4. \quad (18)$$

But the observed dark matter abundance requires a much smaller cold relic density at the 40 MeV epoch. Using

$$\eta_B \sim 6 \times 10^{-10}, \quad s \approx 7.04 n_\gamma, \quad (19)$$

one finds

$$\frac{n_b}{s} \sim 8.5 \times 10^{-11}. \quad (20)$$

The required dark relic energy per entropy is then

$$\frac{\rho_R}{s} \sim 5.3 m_p \frac{n_b}{s} \sim 4 \times 10^{-10} \text{ GeV}. \quad (21)$$

At  $T_c \sim 40 \text{ MeV}$ , the entropy density is roughly

$$s(T_c) = \frac{2\pi^2}{45} g_{*s} T_c^3 \sim 3 \times 10^{-4} - 4 \times 10^{-4} \text{ GeV}^3. \quad (22)$$

Therefore the required cold relic density at that epoch is approximately

$$\rho_R(T_c) \sim 10^{-13} \text{ GeV}^4. \quad (23)$$

The naive one-scale condensate gives  $\rho_\phi \sim 10^{-6} \text{ GeV}^4$ . Thus it overproduces dark matter by roughly

$$\boxed{10^6 - 10^7}. \quad (24)$$

*Reading:* This is the first real quantitative warning. It does not kill the SI relic idea. It does disfavor the simplest version in which the relic mass, relic amplitude, and transition temperature are all set by the same 40 MeV scale.

A viable scalar condensate requires approximately

$$m_\phi \phi_0 \sim 10^{-6} \text{ GeV}^2, \quad (25)$$

not

$$m_\phi \phi_0 \sim T_c^2 \sim 1.6 \times 10^{-3} \text{ GeV}^2. \quad (26)$$

Examples are:

$$\phi_0 \sim 40 \text{ MeV} \quad \Rightarrow \quad m_\phi \sim 10 - 30 \text{ keV}, \quad (27)$$

$$m_\phi \sim 40 \text{ MeV} \quad \Rightarrow \quad \phi_0 \sim 10 - 30 \text{ keV}, \quad (28)$$

$$m_\phi \sim \phi_0 \quad \Rightarrow \quad m_\phi \sim \phi_0 \sim 1 \text{ MeV}. \quad (29)$$

Thus  $T_c \sim 40 \text{ MeV}$  may still set the branch-separation epoch, but it cannot automatically set every relic-sector scale.

## 6 Xenon nuclear recoil as a possible signature class

The xenon recoil discussion should not be built around a claim of discovery. Popular coverage described the LZ event as possible first signs of dark matter, but that is headline-level framing. The LZ result itself reports one event consistent with a nuclear recoil in an extended search window up to about 270 keV, with  $2.6\sigma$  global significance and  $3.4\sigma$  maximum local significance across tested models.

Therefore the conservative interpretation remains:

$$\text{background until repeated and confirmed.} \quad (30)$$

Nevertheless, the event motivates a possible signature class:

$$R_{\text{SI}} + Xe \rightarrow R_{\text{SI}} + Xe. \quad (31)$$

That is, a cold neutral SI relic scatters from a xenon nucleus and produces a nuclear recoil.

This recoil is not an electron recoil. A fast electron can ionize xenon, but it cannot easily transfer enough momentum to the whole xenon nucleus to produce a recoil of hundreds of keV. A heavy neutral projectile, such as a neutron or a sufficiently massive relic, is kinematically more plausible.

That is also why ordinary neutral backgrounds, especially neutron-like backgrounds, must remain the first suspicion.

The SI interpretation becomes interesting only if future events satisfy the following conditions: they repeat; they remain nuclear recoils rather than electron recoils; they have no charged-particle track; they survive neutron-background modeling; they do not come with ordinary electromagnetic shower signatures; and they favor a recoil spectrum that is difficult to explain with standard backgrounds.

The conjecture is therefore not

$$\text{LZ proves SI relics.} \quad (32)$$

It is

$$\text{repeated high-energy xenon nuclear recoils would constrain possible cold SI relics.} \quad (33)$$

## 7 Interaction: rare nuclear portal, not dark gas

If  $R_{\text{SI}}$  can scatter from xenon, it must still avoid obvious astrophysical visibility. This requires a sharp distinction:

$$\text{rare recoil portal} \neq \text{dissipative matter coupling.} \quad (34)$$

The SI relic may occasionally kick a nucleus. It must not behave like gas. If it interacts often enough to cool, lose angular momentum, heat galactic gas, or settle into accretion disks, then it should already be visible through astrophysical distortions.

Therefore the allowed interaction is narrow:

$$\text{rare, non-dissipative nuclear recoil portal.} \quad (35)$$

The allowed process is

$$R_{\text{SI}} + N \rightarrow R_{\text{SI}} + N, \quad (36)$$

where  $N$  is a nucleus.

An inelastic process may also be considered:

$$R_1 + N \rightarrow R_2 + N. \quad (37)$$

But inelastic scattering should not be described as making the event easier. It introduces a threshold cost. Its value is that it may suppress low-energy recoils and select a harder recoil spectrum. In other words, it can reshape the spectrum, but it is not a free rate booster.

Thus a detector-active SI relic is possible only if three requirements are met simultaneously:

$$\text{cold cosmological behavior,} \quad (38)$$

$$\text{rare nuclear interaction,} \quad (39)$$

$$\text{no efficient cooling or gas-like dynamics.} \quad (40)$$

## 8 The detector-active component problem

A smooth scalar condensate can behave as cold dark matter, but it does not automatically produce localized xenon nuclear recoils. A xenon nuclear recoil looks particle-like or lump-like. It requires localized momentum transfer.

Therefore there are two possibilities.

First, the dominant  $R_{\text{SI}}$  dark matter is itself made of localized massive relics. Then the same component supplies both cosmological abundance and detector recoil. This is the cleanest possibility, because the relic-statistics equation directly constrains the detector-active object.

Second, the dominant  $R_{\text{SI}}$  dark matter is a smooth condensate, while only a rare subcomponent appears as compact detector-active excitations. This is possible, but it is less predictive unless the subcomponent abundance, mass, and coupling are derived from the same SI structure.

The second possibility must not be sold as a solution. It is an added burden. A detector-active SI relic requires an additional calculation. If it is not linked to the dominant relic abundance, its mass, number density, and cross-section remain free parameters.

Therefore the preferred version of Conjecture I is not a two-component escape. The preferred version is a single constrained relic population, or a derived relation between the bulk condensate and rare compact excitations.

## 9 What Conjecture I establishes

Conjecture I establishes four points.

First,  $T_c \sim 40 \text{ MeV}$  is not decorative. It is the phase-transition clock inferred from the observed matter/photon residue.

Second,  $R_{\text{SI}}$  cannot be radiation-like if it is to be dark matter. It must be cold, neutral, and matter-like.

Third,

$$m_R n_R \approx 5.3 m_p n_b \quad (41)$$

is the central relic-statistics constraint.

Fourth, the naive one-scale condensate,

$$m_\phi \sim \phi_0 \sim T_c, \quad (42)$$

overcloses by approximately  $10^6$ – $10^7$ .

That is the first quantitative result of the relic program. The consequence is important: the 40 MeV transition may estimate when branch separation occurs, but the dark relic sector cannot be assumed to inherit all of its mass and amplitude scales directly from  $T_c$ . The one-scale unification is therefore under pressure.

## 10 What Conjecture I does not claim

Conjecture I does not claim that LZ discovered dark matter. It does not claim that one xenon event supports the SI framework. It does not claim that arbitrary membrane debris explains dark matter. It does not claim that  $R_{\text{SI}}$  can be dark radiation. It does not claim that a smooth condensate automatically produces nuclear recoils. It does not claim that a detector-active subcomponent is acceptable unless its statistics are derived. It does not claim that all relic-sector scales are naturally equal to 40 MeV.

Its claim is narrower: a cold neutral SI relic sector is a possible dark matter channel only if its relic statistics are derived, and rare xenon nuclear recoils, if repeated and confirmed, may become a laboratory constraint on the detector-active form of that sector.

## 11 Proposed opening outline for Opus IV

Opus IV should begin where Conjecture I stops. It should not be another conceptual restatement. It should be a calculation of the SI relic sector.

The first problem is

$$\kappa_R = \frac{n_R}{n_b}. \quad (43)$$

Once  $\kappa_R$  is known, the relic mass is fixed:

$$m_R \approx \frac{5.3}{\kappa_R} m_p. \quad (44)$$

The second problem is whether the relic is a condensate, compact object, massive particle-like excitation, or soliton. The third problem is whether the same relic can also produce a rare nuclear recoil without becoming astrophysically visible.

A proposed Opus IV structure is:

### 1. Relic statistics from SI projection

Define the relic number density relative to baryons,

$$\kappa_R = n_R/n_b, \quad (45)$$

and attempt to derive it from SI projection multiplicities, index structure, or compact relic formation statistics.

### 2. Condensate test

Develop the scalar-condensate calculation properly, including oscillation onset, entropy dilution,  $g_*(T)$ , and the required relation between  $m_\phi$  and  $\phi_0$ . The starting result is already known:  $m_\phi \sim \phi_0 \sim 40 \text{ MeV}$  overcloses by  $10^6$ – $10^7$ . Opus IV must determine whether this failure can be repaired naturally.



### 3. Compact relic route

If the condensate fails as the dominant dark matter component, consider compact neutral SI relics. The target is not merely stability. The target is the number  $\kappa_R$ .

### 4. Detector portal

If a xenon-recoil signature is retained, define an effective nuclear-recoil portal,

$$R_{\text{SI}} + N \rightarrow R_{\text{SI}} + N. \quad (46)$$

The portal must be rare, non-dissipative, and compatible with gas-cloud, accretion-disk, CMB, BBN, and direct-detection constraints.

### 5. No-go condition

If no candidate can satisfy

$$\rho_R/\rho_b \approx 5.3, \quad R_{\text{SI}} \neq \text{dark radiation}, \quad (47)$$

while also avoiding dissipative gas-like behavior and preserving a rare nuclear portal if detector-active, then the SI dark matter channel fails, even if the branch-relative matter-asymmetry framework remains intact.

## 12 Conclusion

Conjecture I preserves the original Opus III mechanism while making its first phenomenological constraint explicit.

The observed matter/photon ratio gives the empirical estimate of the SI branch-separation epoch:

$$T_c \sim 38\text{--}40 \text{ MeV}. \quad (48)$$

That number is not a relic mass scale by default. It is the timing of the SI condensation event.

The dark matter sector is constrained separately by

$$\boxed{m_R n_R \approx 5.3 m_p n_b}. \quad (49)$$

A naive scalar condensate with every scale set by  $T_c$  fails this test by overproducing dark matter by roughly  $10^6\text{--}10^7$ . Therefore the simplest one-scale relic picture is disfavored.

This is not a defeat of Opus III. It is the first useful pressure test. The viable SI relic must be cold, neutral, nonrelativistic, and statistically constrained. If future xenon-type detectors repeatedly observe high-energy nuclear recoils inconsistent with known backgrounds, such recoils may become a laboratory probe of the detector-active form of  $R_{\text{SI}}$ . Until then, the xenon connection remains conjectural.

The burden of Opus IV is therefore clear:

$$\boxed{\text{derive the relic statistics.}}$$

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